| Question | | Answer | Marks | AO | Guidance |
|---|---|---|---|---|---|
| 11 | (a) | $(x-4)^2+(y+1)^2=13^2$ **oe** | M1 | 1.1 | Correct form; allow either one sign error within brackets or 13 not squared but not both |
| | | all correct **isw** | A1 | 1.1 | |
| | | | [2] | | |
| 11 | (b) | Gradient of radius is $\left[\frac{11--1}{9-4}=\right]\frac{12}{5}$ | B1 | 1.1 | |
| | | Gradient of tangent is $-\frac{5}{12}$ | M1 | 1.1 | **FT** negative reciprocal of their numerical gradient soi |
| | | $y-11=-\frac{5}{12}(x-9)$ **oe** | M1 | 1.1a | $y=-\frac{5}{12}x+c \Rightarrow 11=-\frac{5}{12}\times 9+c \Longrightarrow c=\cdots$ |
| | | $y=-\frac{5}{12}x+\frac{59}{4}$ or $12y+5x=177$ **oe** | A1 | 1.1 | allow any correct simplified form |
| | | | [4] | | |